Effect of Vehicle Speed on Wireless Communication: Doppler Shift and Rayleigh Fading Analysis

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1 Objective

To investigate the effects of vehicle or receiver speed on wireless communication quality, this project studies the impact of Doppler shift and Rayleigh fading on the Bit Error Rate (BER) of Quadrature Phase Shift Keying (QPSK) communication systems.

2 Problem Statement

Mobile Wireless communication systems must contend with a time-varying channel caused by the relative motion between the transmitter and receiver. This motion introduces a Doppler shift in the received signal frequency, and when combined with multipath propagation, gives rise to Rayleigh fading, rapid and random fluctuations in the received signal envelope. As vehicle speed increases, the channel decorrelates more quickly (shorter coherence time), which can degrade the reliability of digital communication systems that rely on accurate channel knowledge at the receiver.

This project investigates the extent to which vehicle speed affects wireless link performance, distinguishing between the fundamental statistical effect of fading itself and the practical effect of a receiver's ability (or inability) to track a fast-changing channel using periodic pilot symbols.

3 Theory and Background

3.1 Doppler Fading

Doppler Fading is a unique aspect of wireless communication systems. The Doppler shift is a fundamental principle related to the electromagnetic radio-wave propagation. The Doppler shift associated with an electromagnetic wave is defined as the perceived change in the frequency of the wave due to relative motion between the transmitter and receiver.

The Doppler shift is given by [1]:

$$f_d = \frac{v}{\lambda} \cos\theta = \frac{v f_c}{c} \cos\theta \tag{1}$$

where θ is the angle of the line joining the mobile receiver and the base station, v is the velocity of the mobile receiver, f_c is the carrier frequency, and c is the speed of light.

3.2 Multipath Propagation and Rayleigh Fading

In a mobile environment without a dominant line-of-sight path, the received signal is the sum of many multipath components arriving from different angles, each with a random amplitude, phase, and Doppler shift. By the Central Limit Theorem, the in-phase and quadrature components of the resulting received signal approach independent, zero-mean Gaussian random processes for a large number of scatterers [1]. The envelope of this process, $|h(t)|$, is Rayleigh distributed:

$$p(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r \geq 0 \tag{2}$$

This model, known as *Clarke's reference model*, forms the theoretical basis for simulating Rayleigh fading channels.

3.3 Jakes Doppler Power Spectrum

The multipath components do not arrive with a single Doppler shift, but rather span the range $[-f_{d,max}, f_{d,max}]$, where $f_{d,max}$ is the maximum frequency deviation. The resulting Doppler power spectral density, known as the Jakes spectrum, is given by:

$$S(f) = \frac{1}{\pi f_{d,max} \sqrt{1 - (f/f_{d,max})^2}}, \quad |f| < f_{d,max} \tag{3}$$

This characteristic "bathtub" shape governs how quickly the fading channel varies in time: a larger $f_{d,max}$ (higher speed) corresponds to a wider spectral spread, and therefore a faster-varying channel envelope.

3.4 Coherence Time

The coherence time T_c is a statistical measure of the time duration over which the channel can be considered approximately constant. A common approximation is [1]:

$$T_c \approx \frac{9}{16\pi f_{d,max}} \tag{4}$$

As speed increases, $f_{d,max}$ increases, and T_c which means that the channel changes more rapidly, and any channel knowledge obtained by the receiver becomes outdated more quickly.

3.5 Rayleigh Fading Channel Simulation Methods

Two independent methods were used in this project to generate a discrete-time, correlated Rayleigh fading channel $h(t)$:

3.5.1 Smith's Spectral Simulation Method

This method shapes complex white Gaussian noise in the frequency domain by the square root of the Jakes power spectrum (Eq. 3), and then applies an inverse FFT to obtain a time-domain fading waveform. This approach is a discrete, computationally efficient realization of the filtering technique originally proposed by Jakes [1] and formalized by Smith.

3.5.2 Zheng–Xiao Sum-of-Sinusoids Method

This method constructs $h(t)$ directly as a sum of a small number (M) of sinusoids with randomized phase, angle of arrival, and initial phase parameters [2]:

$$X_c(t) = \frac{2}{\sqrt{M}} \sum_{n=1}^{M} \cos(\psi_n) \cos(w_d t \cos \alpha_n + \phi) \tag{5}$$

$$X_s(t) = \frac{2}{\sqrt{M}} \sum_{n=1}^{M} \sin(\psi_n) \cos(w_d t \cos \alpha_n + \phi) \tag{6}$$

with $\alpha_n = \frac{2\pi n - \pi + \theta}{4M}$, and θ, ϕ, ψ_n independently and uniformly distributed on $[-\pi, \pi]$. Zheng and Xiao proved that this construction yields exact second-order statistics (autocorrelation) matching the theoretical Rayleigh reference model, even for small M (e.g., $M = 8$), unlike the original deterministic Jakes simulator.

3.6 QPSK Modulation

Quadrature Phase Shift Keying (QPSK) maps pairs of bits to one of four constellation points on the unit circle. With Gray-coded mapping and normalized average symbol energy $E_s = 1$, the transmitted symbol is:

$$s = \frac{1}{\sqrt{2}} \left[(2b_1 - 1) + j(2b_0 - 1)\right] \tag{7}$$

where $b_1, b_0 \in \{0, 1\}$ are the two bits mapped to that symbol.

3.7 Channel Model and Signal-to-Noise Ratio

The received signal after passing through a flat-fading channel with additive white Gaussian noise (AWGN) is modeled as:

$$y(t) = h(t)\, s(t) + n(t) \tag{8}$$

Since QPSK conveys 2 bits per symbol, $E_s = 2E_b$. With $E_s = 1$, the noise spectral density corresponding to a given E_b/N_0 (linear) is:

$$N_0 = \frac{1}{2(E_b/N_0)} \tag{9}$$

3.8 Equalization and Channel State Information (CSI)

To recover the transmitted symbol, the receiver must undo the channel's effect, which requires knowledge of $h(t)$ – referred to as Channel State Information (CSI). Two CSI scenarios were studied:

- **Perfect CSI**: the receiver has exact, instantaneous knowledge of $h(t)$.
- **Estimated CSI**: the receiver estimates $h(t)$ only at known pilot symbol positions and interpolates between them, as is done in practical systems.

For equalization the Zero-Forcing method was implemented:

$$\text{Zero-Forcing (ZF):} \quad \hat{s} = \frac{y}{h} \tag{10}$$

$$\tag{11}$$

ZF perfectly inverts the channel but amplifies noise significantly when $|h|$ is small (deep fade).

3.9 Theoretical QPSK BER over Rayleigh Fading

For QPSK with coherent detection over a Rayleigh fading channel under perfect CSI, the average BER as a function of average E_b/N_0 (γ) has the expression [3]:

$$P_b = \frac{1}{2}\left(1 - \sqrt{\frac{\gamma}{1+\gamma}}\right) \tag{12}$$

This expression is used throughout this report as the reference for validating simulation results.

4 Simulation Methodology

4.1 Software Used

All simulations were implemented in **Python**, using the *NumPy* library for numerical computations, *SciPy* for channel interpolation, and *Matplotlib* for visualization.

4.2 Simulation Parameters

Symbol rate of 10 kHz was chosen to balance computation speed and simulation artifacts due to the Nyquist limit.

Parameter	Value
Carrier frequency, f_c	2 GHz
Vehicle speeds	20, 60, 120 kmph
Symbol rate, f_s	10 kHz
Modulation	QPSK
Number of bits simulated	10^5–10^6
E_b/N_0 sweep range	0–25 dB (2 dB steps)
Zheng–Xiao sinusoid count, M	12
Pilot spacing (sparse case)	20 symbols
Pilot spacing (matched case)	8 symbols
Channel estimation method	Linear interpolation between pilots
Equalization method	Zero-Forcing

Table 1: Simulation parameters used throughout this project.

4.3 Simulation Flow

The simulation proceeds through following stages for each speed and E_b/N_0 value:

1. Generate a random bit-stream and modulate it into QPSK symbols (Eq. 7))
2. Insert known pilot symbols at fixed intervals into the transmitted symbol bit-stream
3. Generate the Rayleigh fading channel $h(t)$ for the given speed, using either Smith's spectral method or Zheng–Xiao sum-of sinusoids method.
4. Pass the transmitted signal through the fading channel and add AWGN corresponding to the target E_b/N_0 (Eq. 8).
5. Estimate the channel at the receiver by interpolating between the pilot symbols.
6. Equalize the received signal using Zero-Forcing (Eqs. 10).
7. Perform threshold-based QPSK demodulation and compute the Bit Error Rate by comparing recovered bits against the original bitstream.

Figure 1 shows the simulation flow.

5 Discussion

5.1 Graphs and Interpretations

5.1.1 Doppler Spectrum and Channel Gain

Figure 2 shows the Jakes Doppler power spectrum (Eq. 3) for the three simulated speeds. As expected, higher speeds produce a wider spectral spread, with the characteristic "bathtub" sharply cutting off at $\pm f_{d,max}$.

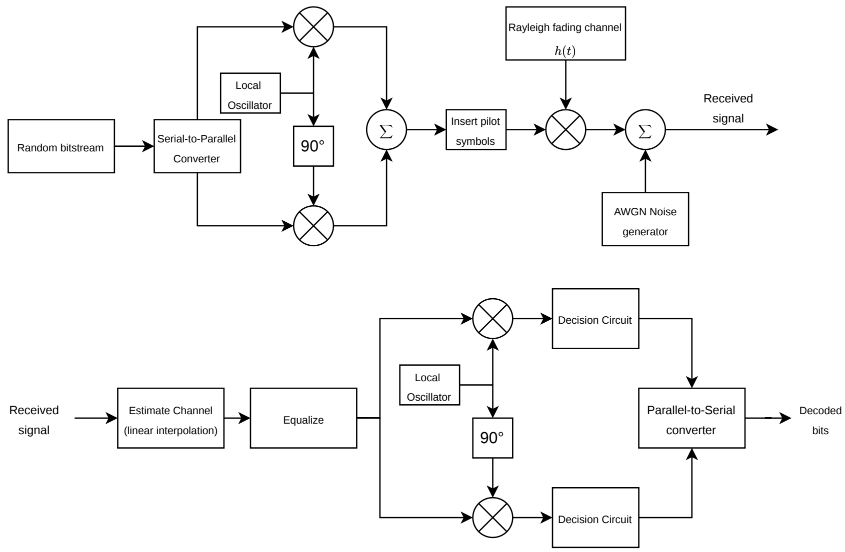


Figure 1: Block diagram of simulation flow.

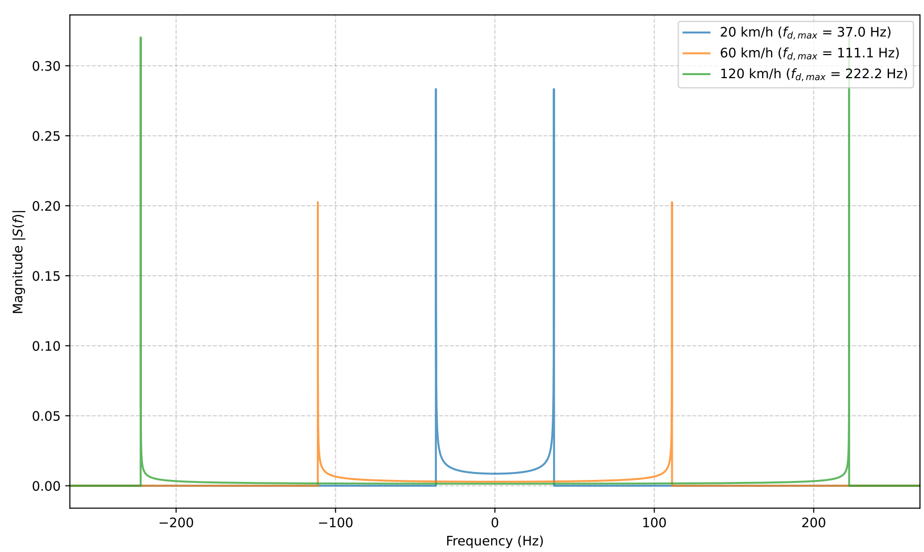


Figure 2: Jakes Doppler Spectrum at 20, 60 and 120 kmph

Figure 3 shows the time-domain fading envelope $|h(t)|$ generated using Smith's spectral method. Faster speeds correspond to more rapid fluctuation of this envelope, reflecting the shorter coherence time associated with higher Doppler spread.

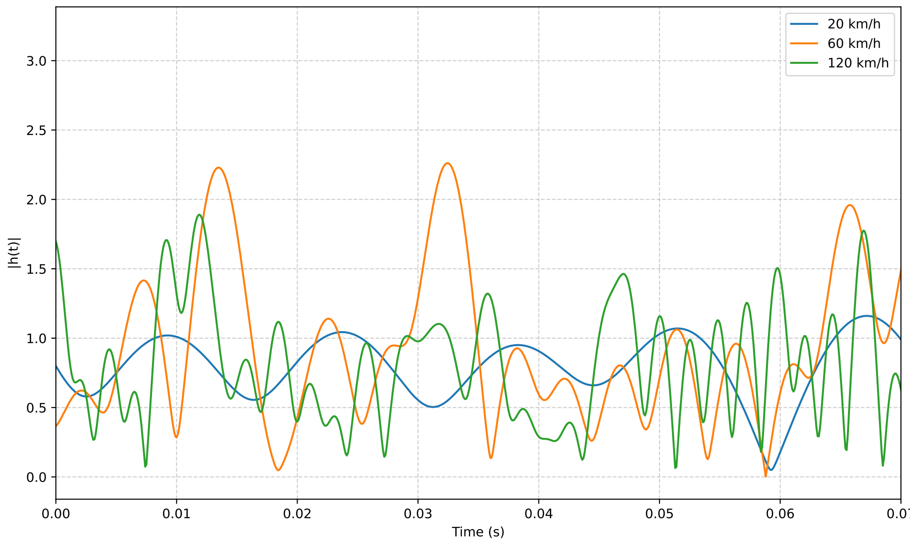


Figure 3: Rayleigh fading envelope $|h(t)|$ vs. time at 20, 60, and 120 kmph.

The statistical validity of the simulated channel was confirmed by comparing the histogram of $|h(t)|$ against the theoretical Rayleigh probability density function (Eq. 2), which showed close agreement for both the Smith and Zheng–Xiao generation methods.

5.1.2 BER under Perfect Channel State Information

Figure 4 shows simulated BER vs. E_b/N_0 under Rayleigh fading, assuming the receiver has perfect, instantaneous knowledge of $h(t)$. All three speeds produce nearly identical BER curves, and both independent fading generation methods (Smith's and Zheng–Xiao) agree closely with the theoretical Rayleigh-fading QPSK BER formula (Eq. 12).

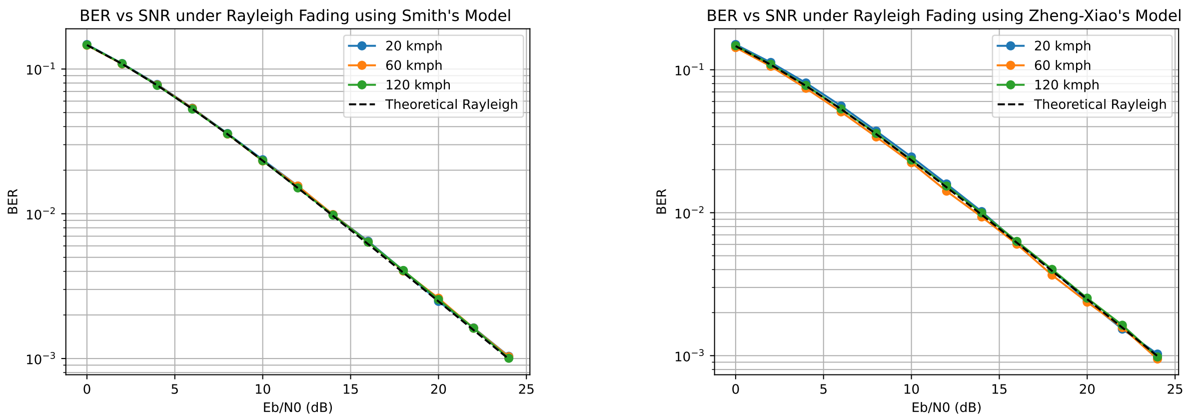
Since the receiver has perfect Channel State Information (CSI) at every instant of time, it instantly knows the exact channel state for every symbol and cancels it out completely.

5.1.3 BER under Realistic Pilot-Based Channel Estimation

Figure 5 shows BER vs. E_b/N_0 when the receiver instead estimates $h(t)$ via linear interpolation between pilot symbols spaced 20 symbols apart. Here, a clear speed-dependent gap emerges: 120 kmph performs consistently worse than 60 kmph, which performs worse than 20 kmph, with the gap widening as E_b/N_0 increases.

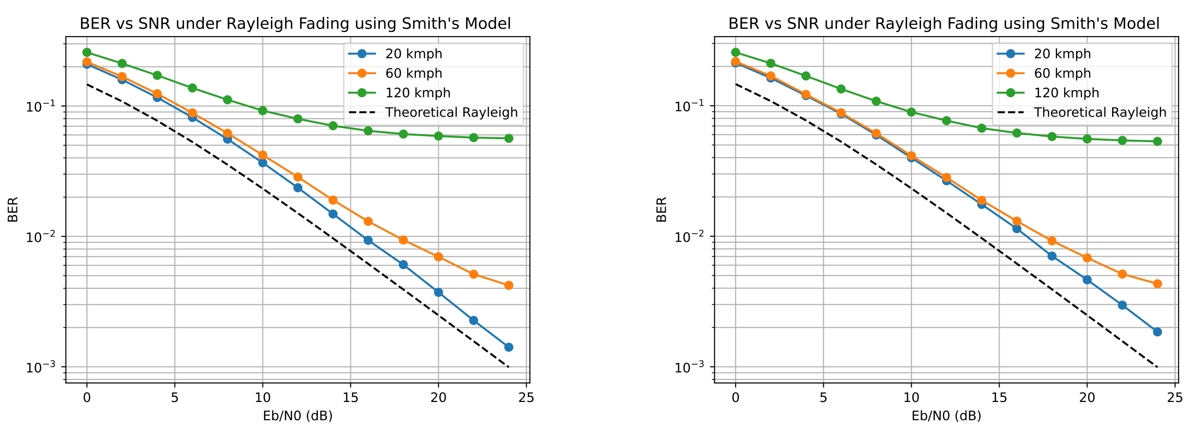
At low SNRs, additive noise is the main cause of bit errors, so speed makes little difference and the curves stay close together.

At high SNRs, noise drops off and channel estimation error takes over as the main bottleneck. Because the channel changes faster at higher speeds, a fixed pilot spacing cannot keep up with rapid fluctuations. This creates an error floor where BER flattens out and stops improving, making the gap between speeds much wider which is clearly visible at 120 kmph.



(a) Simulated using Smith's method (b) Simulated using Zheng-Xiao's method

Figure 4: Simulated BER vs. E_b/N_0 under perfect CSI for 20, 60, and 120 kmph, compared against the theoretical Rayleigh BER curve.

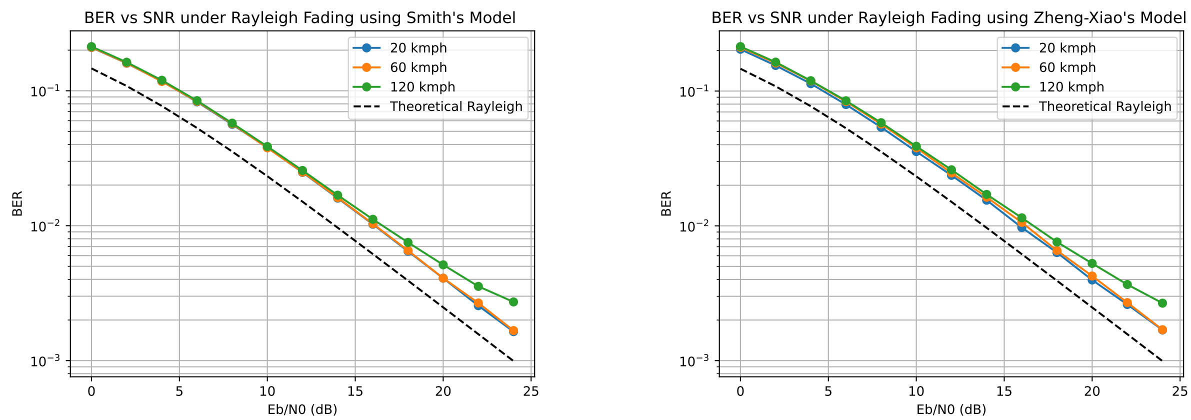


(a) Simulated using Smith's method (b) Simulated using Zheng-Xiao's method

Figure 5: Simulated BER vs. E_b/N_0 with pilot-based channel estimation (pilot spacing = 20 symbols), for 20, 60, and 120 kmph.

5.1.4 Effect of Pilot Spacing

The pilot spacing was reduced to 8 symbols – matching the calculated coherence time at 120 kmph (Eq. 4). Figure 6 shows that under this denser pilot spacing, the BER gap between speeds nearly disappears, with all three curves converging close to the perfect-CSI result.



(a) Simulated using Smith's method

(b) Simulated using Zheng-Xiao method.

Figure 6: Simulated BER vs. E_b/N_0 with pilot spacing reduced to 8 symbols (matched to the 120 kmph coherence time), for 20, 60, and 120 kmph.

This result provides direct, quantitative simulation evidence that speed-dependent BER degradation is driven by *channel estimation staleness* relative to coherence time, rather than by the fading process itself.

6 Conclusion

This project investigated the impact of vehicle speed on wireless communication quality by simulating Doppler shift, Rayleigh fading, and QPSK link performance using two channel generation techniques: Smith's spectral shaping method and the Zheng–Xiao sum-of-sinusoids method. The findings indicate that signal degradation at higher speeds is not driven by fading statistics alone, but by practical channel estimation limits. As vehicle speed increases, the channel's coherence time decreases, causing pilot-based estimates to become stale more quickly. This was demonstrated by observing a significant BER gap between speeds when pilot spacing exceeded the coherence time, and showing that this gap closed when pilot spacing was reduced to match the shortened coherence time at high speeds.

Therefore, at lower speeds the longer coherence time allows pilot-based interpolation to track the channel more accurately which directly results in better performance. At higher speeds, however, the channel changes faster than the receiver can estimate using the pilot-based estimation leading to call drops and reduced communication quality. For this reason, modern cellular standards adaptively scale reference signal (pilot) density for high-mobility user equipment.

Overall, this project confirmed theoretical Rayleigh fading models, explained why vehicle speed degrades wireless signal quality.

References

[1] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[2] Y. R. Zheng and C. Xiao, “Simulation Models With Correct Statistical Properties for Rayleigh Fading Channels,” *IEEE Transactions on Communications*, vol. 51, no. 6, pp. 920–928, Jun. 2003.

[3] M. K. Simon and M.-S. Alouini, *Digital Communication over Fading Channels*, 2nd ed. Hoboken, NJ: Wiley-Interscience, 2005.